Human Faces, Computer Stories: Experimenting with Computer Science Fiction

Abstract:

Computer science continues to redefine our relationship to information, ourselves, and each other. Despite its hold over our lives, its foothold on our literature is relatively small. Literature is meant to document the human experience, but gaps have appeared where it has failed to keep up with technology. For example, a recent article in *MIT Technology Review* exposed machine learning algorithms trained for risk assessment as being biased against marginalized communities. Where are the stories about algorithms inheriting their creators' blind spots? We need new ways of representing computer science in short fiction. I plan on writing a collection of literary short stories which transform computer science concepts into accessible narratives. These stories will be inspired by Italo Calvino's *Cosmicomics*. The *Cosmicomics* provide a foundation upon which I hope to build a unique style of conceptual yet accessible computer science fiction. I will research computer science concepts, analyze the *Cosmicomics*, do exploratory redrafts, then work towards getting the stories published. The redrafting process is key, allowing me to experiment with a variety of themes, literalized metaphors, and points of view. I will assess my final product based on its accessibility, how compelling it is, and whether it meaningfully connects computer science to the human condition.

Objectives and Contribution of the Research:

Computer science has taken over our lives. The themes of computer science in literary short fiction have become increasingly important as the field redefines our relationship to information, ourselves, and each other. By computer science fiction, I mean stories with themes or subjects belonging to the field of computer science. This includes both mechanical devices and theoretical concepts. As a creative writing and information systems double major, I have noticed that the field's real-world importance is not reflected in its literary output or its influence. Now more than ever, we require new ways of representing computer science fiction, so that we can explore with nuance and power the field's impact on humanity.

I plan on writing a collection of literary short stories which transform computer science concepts into accessible narratives. These stories will be inspired by Italo Calvino's *Cosmicomics*, particularly his technique of creating narratives out of scientific concepts. The narratives and conflicts will be character-driven despite being based on scientific concepts.

Twentieth-century Italian author Italo Calvino was famous for portraying reality as if it were a fable. In *The Complete Cosmicomics*, Calvino worked his fantastical style into a science fiction collection. Each story takes a natural phenomenon such as the speed of light and turns it into a fantastical narrative. This allows for abstract scientific truths to be translated into literalized metaphors and emotional truths. Literalized metaphors are concepts which are literally transformed into plots, characters, or settings. For example, exploring the topic of hospital burnout by having the hospital staff transform into literal zombies. This metaphor operates on an emotional logic—the comparison feels emotionally true. Calvino's *Cosmicomics* are novel

because they combine emotional truths with technical concepts. “At once experimental and accessible” (Ricci 6), the *Cosmicomics* provide a foundation upon which I can write a new type of computer science short fiction that is more accessible, efficient, and flexible.

While successful computer science short stories do exist, there are few obvious solutions when it comes to writing one. One of the best ways of representing abstract, scientific concepts is hard science fiction. Charles Stross’s *Accelerando* and Greg Egan’s “Dust” show that while hard science fiction allows you to explore computer science themes in depth, it risks making your story dense and unapproachable. On the other hand, non-technical computer science fiction often limits itself to well-known computer science subjects: computers (Dick’s “The Great C”), artificial intelligence (Vonnegut’s “EPICAC”), and virtual reality (Bradbury’s “The Veldt”). While substantial, these stories often avoid more conceptual, domain-specific concepts. Ted Chiang’s novella *The Lifecycle of Software Objects* is a good example of what these stories miss out on. Its subject is the Software Development Life Cycle, an organizational technique most readers would be unfamiliar with. By writing in the *Cosmicomics*’ tradition, I hope to produce stories which are both conceptual and accessible.

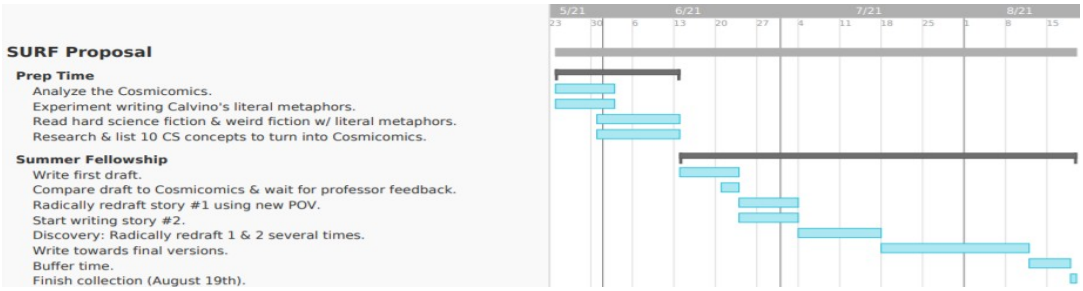
Last semester, I took Writing and Reading Science Fiction at the University of Iowa. Professor Ren Arcamone stressed that short science fiction is hard because it requires worldbuilding, which is difficult to accomplish within the space of a short story. The *Cosmicomics*’ fantastical narratives playfully subordinate setting, character, and point of view to the story’s core concept. As a consequence, Calvino’s worldbuilding actually serves the greater narrative rather than demanding things from it. Stories centered around a technical concept must also explain that concept without boring the reader, losing them, or using too much text. The *Cosmicomics* model combats this by focusing entirely on concept and closely coupling exposition and plot. As an example, “The Light-Years” embeds exposition into a plot twist.

I hypothesize that computer science fiction is often more difficult to represent than traditional science fiction because computers are unobservable, unintuitive, and highly abstract. Traditional science fiction is grounded in the natural sciences and closer to our own experience with nature—you can visualize the orbit of a planet. This makes the natural sciences simpler for readers to understand, and easier to transform into narrative components like setting, plots, or character. Using *Cosmicomics*’ literalized metaphors will connect readers to abstract, unintuitive concepts. As an example, “All at One Point” is about a community who all share a single point in space. Unlike realism, Calvino’s fantastical realities can change shape to help represent concepts.

Methodology:

I am writing a collection of literary short stories because it is the writer’s responsibility to successfully communicate themes and subjects. Experimenting with short stories will prove more fruitful than with novels because their limited length breeds innovation. Science fiction is the obvious choice for genre because of its sharp focus on technical subjects. In an interview with *The Believer*, science fiction author Ted Chiang suggests science fiction brings readers closer to abstract and intellectual concepts, offering them a chance to get “genuinely invested in the outcome of a thought experiment.” In his book *Metamorphoses of Science Fiction*, Darko Suvin

identifies science fiction as a nonmimetic genre, meaning that readers of science fiction do not expect its stories to recreate everyday reality. For literalized metaphors to work, readers must be willing to suspend their disbelief and follow along with imagined logic.



Project timeline, May to August 2021

Prep work will be done before summer research starts. I will analyze the *Cosmicomics* by looking at structure, how Calvino develops his themes, and how he connects his characters and plots to the original theoretical concept. I will experiment with Calvino's style by using a technique I was taught at the University of Iowa. It involves breaking down an author's style into pattern, syntax, and diction, then imitating key aspects. The redrafting process will be highly exploratory because the *Cosmicomics* format is very flexible. It will likely take several attempts to find the best way to represent these concepts. I will try different points of view, create different literalized metaphors, take stories in different thematic directions, and add or remove characters. Finally, I believe a long revision process is necessary for good writing—it took author Hubert Selby Jr. six years of redrafting to submit his first book of short stories. Buffer time, precedent, and the fact I can reduce scope to one short story make this project feasible.

Here are my expected creative outcomes. Each story needs to explore a computer science concept. Each story should be accessible, so that a general audience with no prior knowledge can enter the text and understand it. Artistically, I hope to better understand how to create literalized metaphors, how to embed exposition into narrative, and what separates computer science fiction from traditional science fiction.

Background:

I have experience transforming computer science concepts into fiction. Like Calvino, my writing style is playful, experimental, and grounded in literal metaphors. I have known Professor Sharon Dilworth, the head of my department, for a while, and am taking her Survey of Forms: Screenwriting. I asked her to mentor me because of her astute insights and careful feedback.

Feedback and Evaluation:

I will meet with Professor Dilworth weekly, during which time she will give me feedback on my drafts to help rewrite them. Here are my assessment criteria: is the story compelling, is it well-paced, does it convey a computer science concept, does the computer science meaningfully comment on the human condition, and how accessible is the story to a general audience?

Dissemination of Knowledge:

I will submit my short stories to literary journals such as *Rainy Day*, *The Archive*, and *The Foundationalist*. I will also present my findings and my own collection at Meeting of the Minds.

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